

Pre-incisional antibiotic timing and operative case duration in elective neurosurgery: a retrospective EHR cohort study using de-identified UCSF Epic data

Motivation and Background

Perioperative antibiotic prophylaxis aims to help patients achieve adequate tissue concentrations by incision time to reduce postoperative infection. Although surgical-site infections (SSIs) in neurosurgery are relatively rare [1], they are associated with significant morbidity and cost. Current guidelines recommend administration within 60 minutes before incision, or within 120 minutes for agents such as vancomycin for longer infusion time, timing in practice is highly variable and often poorly documented. Using de-identified UCSF Epic data, we evaluated whether peri-incisional antibiotic administration (0–30 minutes before incision) is associated with differences in operative duration and postoperative infection diagnoses compared with earlier guideline-concordant timing (31–120 minutes). This question is clinically relevant because prophylactic timing may reflect perioperative workflow quality and influence downstream surgical outcomes through tissue antibiotic levels, infection risk, and the need for additional procedures.

Study Design

This was a retrospective cohort study using de-identified UCSF Epic clinical data from 2020 through 2022. The analytic unit was the surgical case. The study was designed to identify elective neurosurgical procedures with documented pre-incisional prophylactic cefazolin or vancomycin and compare outcomes across timing groups.

Target population: adults undergoing elective neurosurgical procedures.

Accessible population: adults in the de-identified UCSF Epic data warehouse with elective neurosurgical surgical cases between January 1, 2020 and December 31, 2022, documented procedure start times, and a pre-incisional cefazolin or vancomycin administration within 120 minutes before incision.

Inclusion criteria

- Surgical case date between 2020-01-01 and 2022-12-31
- Elective surgery (`Classification = 'F-Elective'`)
- Neurosurgical service identified through `SurgicalProcedureEventFact (Service = 'Neurological Surgery')`
- Age 18–100 years at surgery
- Valid `PatientDurableKey`
- Non-missing `ProcedureStartInstant`
- At least one qualifying intravenous cefazolin or vancomycin administration before incision

- Final pre-incisional prophylactic administration occurring 0–120 minutes before incision

Exclusion criteria

- Missing or invalid patient identifier
- Missing procedure start time
- No qualifying pre-incisional cefazolin or vancomycin administration within 120 minutes before incision
- Age under 18 years or over 100 years

Cohort

Both inpatient and outpatient cases (`AdmissionPatientClass`) were included to reflect real-world practice. Restricting the cohort to elective cases helped reduce emergency-related variability. For each surgical case, we selected the *last documented antibiotic administration before incision*.

Predictor

The main predictor was timing of the final pre-incisional prophylactic antibiotic administration relative to incision

Primary exposure grouping (2 groups)

- Peri-incisional: 0–30 minutes before incision → antibiotic may not have reached tissues yet
- Early prophylaxis: 31–120 minutes before incision → guideline concordant

Secondary exposure grouping (3 groups)

- Peri-incisional: 0–30 minutes before incision
- Optimal prophylaxis: 31–60 minutes before incision → guideline target
- 61–120 minutes before incision → guideline acceptable, may be too early

Outcomes

Primary outcome: Operative case duration (minutes), defined as the difference in minutes between `ProcedureStartInstant` and `ProcedureCompleteInstant`.

Secondary outcomes

1. 30-day postoperative infection diagnosis, defined using selected ICD-10-CM codes within 30 days after surgery (T81.4x, L08.9, A41.x, T85.7x)
2. Length of stay in days (LOS), restricted to inpatient cases only

*We initially considered 30-day postoperative infection as the primary outcome, but events were uncommon and the analysis was underpowered (26 total events in the cohort). Therefore, we chose operative case duration, a more reliably and clinically relevant time-based measure, as the primary outcome. 30-day postoperative infection was retained as a secondary outcome.

Covariates

- **Demographics:** age, sex, race/ethnicity
- **Comorbidities (12-month lookback):** diabetes, hypertension, obesity, smoking, cancer
- **Laboratory values:** most recent preoperative WBC and hemoglobin within 30 days before surgery
- **Medication-related variables:** antibiotic formulation/name and antibiotic class (cefazolin vs vancomycin)
- **Encounter characteristic:** admission patient class
- **Provider characteristic:** surgical/encounter provider specialty or attending specialty
- **Vital sign:** preop heart rate, if available in the structured EHR vitals/flowsheet table

Because EHR fields are heterogeneous, candidate medications and labs were first explored using broader searches and then narrowed to final definitions. Final antibiotic definitions were restricted to intravenous cefazolin and vancomycin formulations, and final lab definitions were restricted to curated blood-based WBC and hemoglobin component keys. Comorbidities were initially defined using ICD-10-CM code groupings (i.e. ‘E09%’, ‘E10%’, etc.) [2] and were subsequently checked against diagnosis groupers identified in `DiagnosisSetDim`, through their `ValueSetEpicId` values (EDG CONCEPT DIABETES MELLITUS, EDG CONCEPT HYPERTENSION, EDG CONCEPT OBESITY, EDG CONCEPT HP TOBACCO USE, and EDG CONCEPT CANCER). The Table 1 values were approximately the same for both.

Characteristic	Peri-Incisional Prophylaxis (0-30 min) n = 566	Early Prophylaxis (31-120 min) n = 260
Age, mean (SD)	56.11 (17.23)	54.72 (16.36)
Sex		
Female	276 (48.8)	130 (50.0)
Male	290 (51.2)	130 (50.0)
Race/Ethnicity		
White	348 (61.5)	155 (59.6)
Asian	72 (12.7)	30 (11.5)
Black or African American	18 (3.2)	11 (4.2)
Latinx	96 (17.0)	37 (14.2)
Multi-Race/Ethnicity	11 (1.9)	5 (1.9)
Native American or Alaska Native	3 (0.5)	3 (1.2)
Native Hawaiian or Other Pacific Islander	4 (0.7)	1 (0.4)
Other	5 (0.9)	5 (1.9)
Southwest Asian and North African	7 (1.2)	8 (3.1)
Unknown/Declined	2 (0.4)	5 (1.9)
Attending Provider Specialty		
Neurosurgery	540 (95.4)	247 (95.0)
Anesthesiology	8 (1.4)	2 (0.8)
Medicine	10 (1.8)	3 (1.2)
Other	8 (1.4)	8 (3.1)
Diabetes		
No	521 (92.0)	243 (93.5)
Yes	45 (8.0)	17 (6.5)
Hypertension		
No	433 (76.5)	212 (81.5)
Yes	133 (23.5)	48 (18.5)
Obesity		
No	547 (96.6)	247 (95.0)
Yes	19 (3.4)	13 (5.0)
Smoking		
No	511 (90.3)	227 (87.3)
Yes	55 (9.7)	33 (12.7)
Cancer		
No	366 (64.7)	166 (63.8)
Yes	200 (35.3)	94 (36.2)
Preoperative WBC, median [IQR]	7.00 [5.68, 9.10]	6.60 [5.40, 8.70]
WBC Category		
Normal (<10)	364 (64.3)	174 (66.9)
Elevated (≥10)	76 (13.4)	31 (11.9)
Missing	126 (22.3)	55 (21.2)
Preoperative Hemoglobin, median [IQR]	13.80 [12.78, 14.70]	13.60 [12.50, 14.80]
Hemoglobin Category		
Normal	349 (61.7)	166 (63.8)
Anemia	91 (16.1)	39 (15.0)
Missing	126 (22.3)	55 (21.2)
Case Duration (min), mean (SD)	171.50 (113.58)	215.22 (111.83)
Antibiotic Agent		
Cefazolin 1 g injection	253 (44.7)	95 (36.5)
Cefazolin 500 mg injection	3 (0.5)	2 (0.8)
Vancomycin 1,000 mg IV injection	25 (4.4)	45 (17.3)
Vancomycin 500 mg IV solution	285 (50.4)	118 (45.4)
Minutes Before Incision, mean (SD)	9.30 (9.44)	52.26 (19.05)

Abbreviations: IQR, interquartile range; SD, standard deviation; WBC, white blood cell count.

Categorical variables reported as n (%). Continuous variables reported as mean (SD) or median [IQR] as appropriate.

Missing values reported where applicable. Elevated WBC defined as $\geq 10 \times 10^9/L$; anemia defined per institutional thresholds.

Table 1. Cohort Characteristics Stratified by Primary Outcome Group

Statistical Analysis

Table 1 summarizes baseline characteristics by primary exposure group (0–30 minutes vs 31–120 minutes before incision). Continuous variables were summarized using medians and interquartile ranges when skewed. Because operative case duration and LOS were not normally distributed, two-group comparisons used the Wilcoxon rank-sum test and three-group sensitivity analyses used the *Kruskal-Wallis* test. Because postoperative infection events were uncommon, infection comparisons used *Fisher's Exact* test.

Results

The final analytic cohort included 826 elective adult neurosurgical cases (566 peri-incisional, 260 earlier-timing). Operative case duration differed significantly between timing groups in both the two-group and three-group analyses (**Wilcoxon $p < 0.001$; Kruskal-Wallis $p < 0.001$**). In contrast, 30-day postoperative infection was uncommon and did not differ significantly (**Fisher $p = 0.399$**), and inpatient length of stay was similar across groups (**Wilcoxon $p = 0.838$; Kruskal-Wallis $p = 0.238$**).

Interestingly, earlier prophylaxis (31–120 minutes) was associated with longer operative duration than peri-incisional (0–30 minute) dosing. This is unlikely to represent a causal effect of timing on surgical duration. Instead, this pattern likely reflects unmeasured confounding, such as case complexity and perioperative workflow. For example, longer or more complex cases, and those receiving vancomycin, may be likely to receive antibiotics earlier. Thus, timing may act more as a marker of underlying complexity rather than a direct determinant of operation time. The main results and secondary outcomes are summarized in Table 2 and Table 3 below.

The study has several limitations. It relied on structured de-identified EHR data, which may be subject to misclassification and incomplete capture of clinical concepts. The postoperative infection outcome was broader than strict surgical site infection because strict SSI counts were too low for stable analysis. We considered only the final pre-incisional dose of cefazolin or vancomycin and did not explicitly consider multiple dosing or redosing patterns. We also did not adjust for the class of antibiotics, even though vancomycin is often administered earlier than cefazolin because of infusion time. Finally, as an observational study, this analysis cannot establish causality, and residual confounding by unmeasured surgical complexity likely influences the observed association between antibiotic timing and operative duration.

*Results were similar after excluding an extreme operative-duration outlier, suggesting that the association was not driven by a single case.

Table 2	Peri-incisional 0–30 min (n = 566)	Earlier timing 31–120 min (n = 260)	p-value
Primary outcome			
Operative case duration (min)	162.5 (71.3–240.0)	212.0 (142.0–277.0)	<0.001 ^a
Secondary outcomes			
30-day postoperative infection	20/566 (3.5%)	6/260 (2.3%)	0.399 ^b
LOS for inpatients (days)	3 (2–7) (n = 455)	3 (2–7) (n = 253)	0.838 ^c

^a Wilcoxon rank-sum test ($W = 54883$, $p = 4.33 \times 10^{-9}$).

^b Fisher's exact test.

^c Wilcoxon rank-sum test.

Table 3	0–30 min (n = 566)	31–60 min (n = 194)	61–120 min (n = 66)	p-value
Primary outcome				
Operative case duration (min)	162.5 (71.3–240.0)	204.5 (140.0–272.3)	234.5 (150.8–284.0)	<0.001 ^a
Secondary outcomes				
30-day postoperative infection	20/566 (3.5%)	5/194 (2.6%)	1/66 (1.5%)	0.758 ^b
LOS for inpatients (days)	3 (2–7) (n = 455)	3 (2–7) (n = 188)	3 (2–8) (n = 65)	0.238 ^c

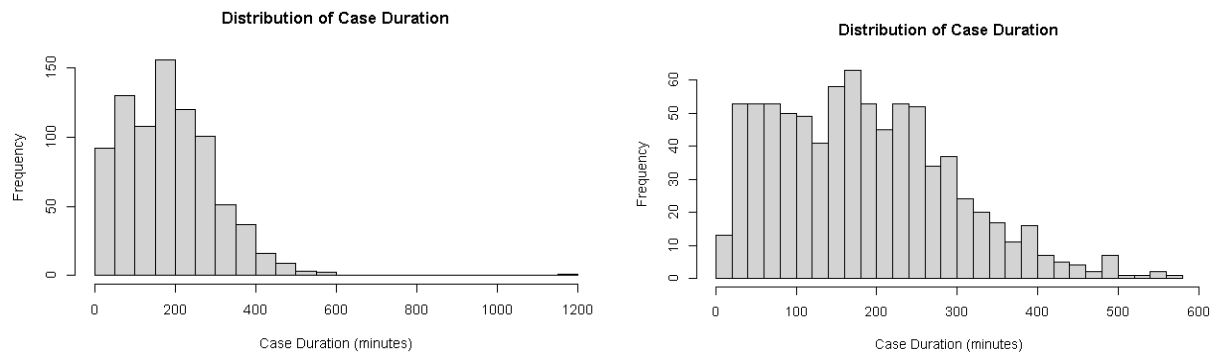
^a Kruskal–Wallis test ($\chi^2 = 36.10$, $df = 2$, $p = 1.45 \times 10^{-8}$). Results were similar after excluding one extreme outlier.

^b Fisher's exact test across three groups.

^c Kruskal–Wallis test ($\chi^2 = 2.87$, $df = 2$)

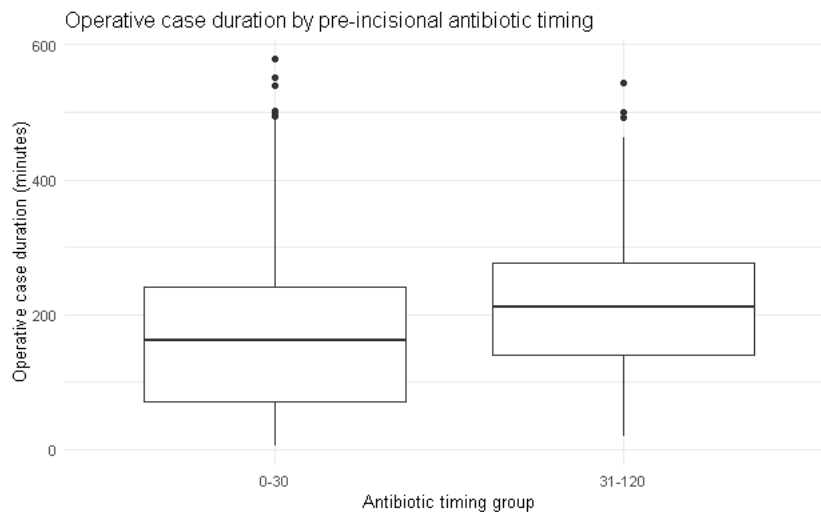
Supplementary Visuals + Analysis

Distribution of Case Duration (Primary Outcome) with vs. without extreme outlier



The distribution is skewed which is why we choose Wilcoxon rank sum test Kruskal-Wallis for continuous outcomes.

Boxplot of Case Durations (Primary Outcome) with removed outlier across Two timing groups



Because event counts were low and the comparison was categorical, I used Fisher's exact test for the 30-day postoperative infection outcome (secondary outcome).

Regression Analysis

$\log(\text{case_duration_min}) \sim \text{timing_group_main} + \text{antibiotic_class} + \text{Age} + \text{Sex} + \text{diabetes}$

Coefficients:

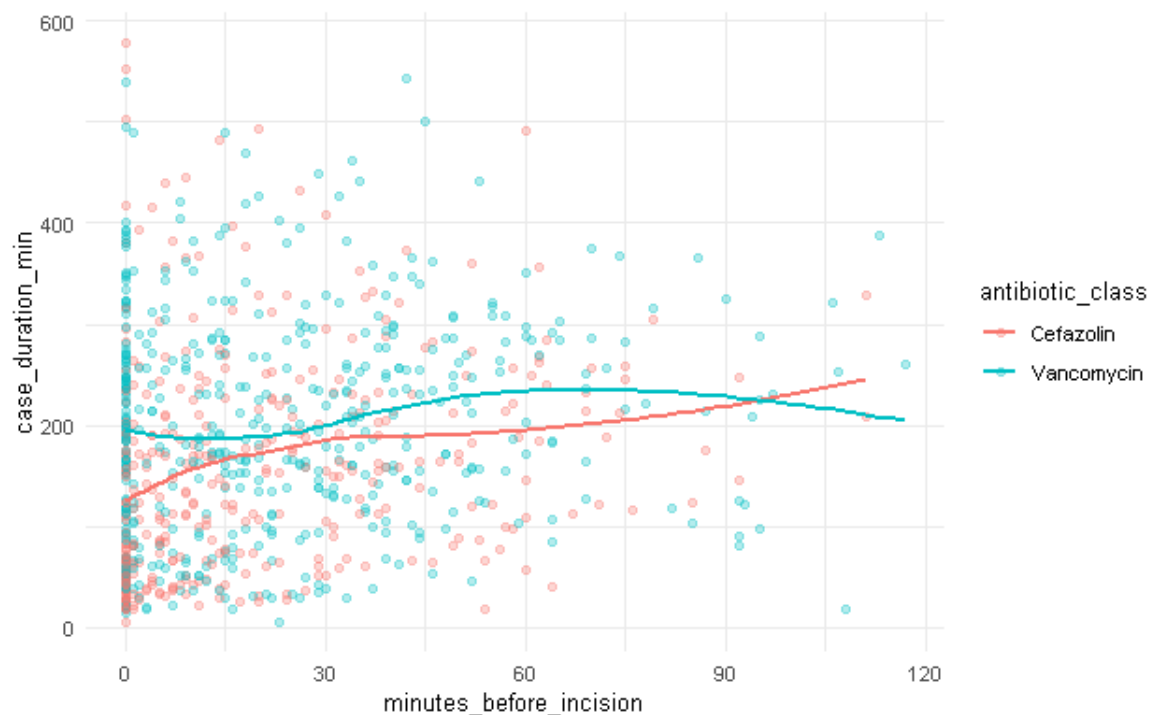
	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	5.010299	0.095095	52.687	< 2e-16	***
timing_group_main31-120	0.335930	0.054644	6.148	1.23e-09	***
antibiotic_classVancomycin	0.294171	0.051273	5.737	1.35e-08	***
Age	-0.005237	0.001507	-3.475	0.000538	***
SexMale	-0.012660	0.050759	-0.249	0.803103	
diabetes	0.003110	0.096968	0.032	0.974426	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7266 on 820 degrees of freedom
Multiple R-squared: 0.09842, Adjusted R-squared: 0.09292
F-statistic: 17.9 on 5 and 820 DF, p-value: < 2.2e-16

We fit a multivariable linear regression model of log-transformed operative duration adjusting for timing group, antibiotic class, age, sex, and diabetes to account for potential confounding. In this model, earlier timing (31–120 vs 0–30 minutes) remained associated with longer operative duration after adjustment. The pre-incision timing group, antibiotic class, and age (negative association) were all statistically significantly associated with duration.

Scatterplot of antibiotic timing (pre-incision) vs. case duration.



There appears to be a weak positive relationship